

**ISIMM – Higher Institute of Computer Science
and Mathematics of Monastir**

Emotion Recognition through Textual Data

Mid-Project Report

Authors:

Khadija Elloumi

Rania Ouselati

Asma Sebai

Nada Aouiti

Date: November 2025

Contents

1	Business Understanding and Problem Formulation	3
1.1	Problem Description	3
1.2	Context and Applications	3
1.3	Motivation	3
1.4	Relation to Machine Learning	4
1.5	Example	4
2	Data Collection	5
2.1	Methodology 1: Automated Tweet Extraction using Selenium	5
2.1.1	Objective	5
2.1.2	Principle	5
2.1.3	Process Steps	5
2.1.4	Expected Outcome	6
2.2	Methodology 2: Reddit Scraping and Synthetic Data Generation	6
2.2.1	Phase 1: Real Data Collection from Reddit	6
2.2.2	Phase 2: Synthetic Data Generation	6
3	Data Understanding	7
3.1	Dataset Description	7
3.2	Dataset Structure	7
3.3	Content Characteristics	8
3.3.1	Text Variety	8
3.3.2	Linguistic Properties	8
3.4	Emotion Label Distribution	8
3.5	Data Quality Notes	9
3.6	Suitability for Machine Learning	9
4	Data Cleaning and Preparation	10
4.1	Introduction	10
4.2	Initial Cleaning	10
4.3	Text Preprocessing Pipeline	10

4.4	Batch Processing	11
4.5	Summary of Cleaning Results	11
4.6	Examples Before and After Preprocessing	12
4.7	Conclusion	12
5	Data Visualization	13
5.1	Visualization Before Preprocessing	13
6	References	15

Chapter 1

Business Understanding and Problem Formulation

1.1 Problem Description

We aim to predict people’s emotional states based on what they say or write. This involves analyzing linguistic patterns within textual data to classify emotions such as sadness, joy, anger, fear, and neutrality.

1.2 Context and Applications

This problem has broad and impactful applications:

- In the medical field, for mental-health clinicians.
- Among family members seeking to understand each other’s emotional states.
- For analyzing complex characters in films, literature, or historical contexts.

Our primary target use case focuses on families—helping members recognize when someone might need emotional or psychological support.

1.3 Motivation

Family members often want to know if a loved one shows early signs of mental health distress. Given a text message or journal entry, our model should output the corresponding emotional state, assisting in detecting early symptoms of depression.

1.4 Relation to Machine Learning

This is a text classification problem within the domain of Natural Language Processing (NLP). The system learns patterns statistically from text labeled with emotions rather than relying on rigid rule-based systems. Machine learning models can generalize from training examples to predict emotional categories such as joy, sadness, fear, or anger.

1.5 Example

For instance, expressions of anger in one culture might be interpreted merely as high energy or loudness in another. Machine learning allows such nuances to be learned empirically from data rather than predefined manually.

Chapter 2

Data Collection

2.1 Methodology 1: Automated Tweet Extraction using Selenium

This method automates the extraction of Arabic-language tweets expressing various emotions such as sadness, joy, fear, anger, disgust, and neutrality.

2.1.1 Objective

The goal is to construct a labeled emotional dataset from publicly available tweets to be used for NLP or supervised learning tasks.

2.1.2 Principle

The process uses Selenium WebDriver to simulate human browsing actions on Twitter. Unlike official APIs, this approach accesses public tweets directly through the web interface, allowing more flexibility.

2.1.3 Process Steps

1. Configure an automated Chrome browser with a dedicated profile.
2. Define keywords representing each emotion in Arabic (e.g., "خوف", "فرح", "حزين").
3. Use Selenium to perform searches and scroll through pages to collect tweets.
4. Detect and label dominant emotions automatically using keyword matching.
5. Save structured results into CSV and JSON files.

2.1.4 Expected Outcome

A structured dataset of several hundred Arabic tweets annotated by emotion, ready for use in NLP tasks.

2.2 Methodology 2: Reddit Scraping and Synthetic Data Generation

This second data collection approach consists of two phases:

2.2.1 Phase 1: Real Data Collection from Reddit

Using Reddit’s public endpoints, posts and comments are retrieved from Arabic-speaking subreddits such as *r/arabs*, *r/Egypt*, and *r/SaudiArabia*. The process includes:

- Filtering text written in Arabic.
- Cleaning non-Arabic characters and redundant spaces.
- Automatically labeling emotions using keyword-based detection.

Each record includes metadata such as upvotes, timestamps, and the Arabic ratio of the text.

2.2.2 Phase 2: Synthetic Data Generation

To complement real data, additional examples are generated using predefined linguistic templates filled with emotion-specific expressions. This ensures dataset balance and increased diversity.

Example template:

أفعل ماذا أعرف ولا الشديد بالحزن أشعر

(“I feel deep sadness and I don’t know what to do; I see no solution.”)

The final dataset includes approximately 2000 samples: 487 real and 1513 synthetic.

Chapter 3

Data Understanding

3.1 Dataset Description

The dataset consists of Arabic text entries collected from online platforms and social media. Each entry includes a short user-generated message alongside a manually assigned emotion label. The dataset is designed to support the development and evaluation of a machine-learning model for emotion classification in Arabic text.

3.2 Dataset Structure

Each record contains three fields:

- **ID:** A numerical or alphanumeric identifier.
- **Text:** An Arabic sentence or short paragraph written in natural, informal language.
- **Emotion Label:** The target class representing the dominant emotion expressed in the text.

The dataset includes the following emotion categories:

- Neutral
- Happiness
- Fear
- Anxiety
- Disgust

These categories reflect a mix of basic emotions observed in psychological literature and commonly used in sentiment classification tasks.

3.3 Content Characteristics

3.3.1 Text Variety

The dataset contains a wide range of text types, including:

- Questions and personal reflections
- Social commentary
- Short emotional statements
- Casual conversational messages
- Requests, complaints, and reactions
- User opinions and reactions to events or news
- Informal dialectal Arabic (Tunisian, Egyptian, Gulf, Levantine, Iraqi...)

Messages range from very short (10–20 characters) to long informal posts exceeding 300 characters.

3.3.2 Linguistic Properties

- Texts are written in Arabic script, including Modern Standard Arabic and multiple dialects.
- Many entries contain colloquial expressions, emojis, or spoken-language structures.
- Grammar, spelling, and diacritics are inconsistent, reflecting real-world user-generated content.
- Some texts include foreign words, transliterations, or mixed language.

This diversity makes the dataset suitable for robust real-world emotion detection.

3.4 Emotion Label Distribution

Although exact counts require formal computation, manual inspection shows:

- Neutral is the largest class.
- Happiness appears in multiple short supportive comments.
- Disgust includes posts containing words like “قرف”.

- Fear appears in reactions expressing fright or concern.
- Anxiety appears in text discussing stress, pressure, or overwhelming situations.

This natural imbalance reflects typical distributions in Arabic social media data, where neutral or informational messages dominate.

3.5 Data Quality Notes

- Some entries appear duplicated (e.g., several repeated ”قرف“ messages).
- Some records contain very long texts, requiring preprocessing such as truncation or normalization.
- Emojis and punctuation vary widely and may need cleaning.
- Some IDs are missing or repeated across sources.
- Emotion labels appear manually assigned and semantically consistent.

3.6 Suitability for Machine Learning

This dataset is well-suited for:

- Text classification
- Emotion recognition in Arabic
- Social media sentiment analysis
- Dialectal Arabic NLP studies

The presence of diverse dialects, real-world noise, and informal language gives your model strong generalization potential.

Chapter 4

Data Cleaning and Preparation

4.1 Introduction

Before training any machine learning model, it is crucial to clean and preprocess the dataset to ensure data quality and consistency. For the Arabic emotion dataset, a comprehensive preprocessing pipeline was applied to handle noise, duplicates, empty fields, and informal text patterns such as emojis, dialectal expressions, and inconsistent spelling.

4.2 Initial Cleaning

The first steps in data cleaning included:

- Filling missing content with empty strings.
- Normalizing emotion labels by converting them to lowercase and stripping whitespace.
- Removing rows with irrelevant labels (e.g., “surprise” or unrelated text entries).
- Dropping duplicate entries to ensure unique texts.
- Removing texts that were empty, contained only numbers, or were not predominantly Arabic.

These steps reduced the dataset by removing invalid or duplicate entries while retaining meaningful Arabic text.

4.3 Text Preprocessing Pipeline

The preprocessing pipeline applied to each text included the following steps:

1. **Validation of Arabic content:** Texts were checked to ensure at least 60% of characters were Arabic.

2. **Normalization:** Removing tatweel (character elongation), extra spaces, and unwanted line breaks.
3. **Emoji handling:** Emojis were replaced with corresponding sentiment words using a custom Arabic emoji-sentiment dictionary.
4. **Cleaning URLs, mentions, and hashtags:** URLs, social media mentions, and hashtags were removed.
5. **Repeated character reduction:** Excessively repeated characters were limited to improve consistency.
6. **Tokenization and lemmatization:** Arabic tokenization and lemmatization were performed using Stanza, a state-of-the-art NLP library.
7. **Stopwords removal:** Common Arabic stopwords were removed to focus on meaningful tokens.
8. **Optional spell correction:** Words were optionally corrected using an Arabic spell checker, primarily for longer words.
9. **Final filtering:** Tokens that were too short or non-Arabic were removed, and texts were reconstructed into clean, normalized sentences.

4.4 Batch Processing

To handle large datasets efficiently, preprocessing was applied in batches of 1000 texts, ensuring duplicates and empty fields were handled before processing each batch. This method allowed scalable and reproducible text cleaning.

4.5 Summary of Cleaning Results

After applying the preprocessing pipeline:

- Initial number of texts: **8331**
- Texts removed due to emptiness or duplication: **5098**
- Final number of cleaned texts: **3233**
- Average text length after cleaning: **20** characters
- Minimum text length: **3** characters
- Maximum text length: **290** characters

4.6 Examples Before and After Preprocessing

- **Original:** رائع! وهذا □ □ اليوم جدا سعيد "أنا"
- **Cleaned:** رائع "اليوم" سعيد
- **Original:** "كدا بحس ليه عارف" مش
- **Cleaned:** "حاسس"

4.7 Conclusion

The dataset is now normalized, free of duplicates, and preprocessed for Arabic-specific linguistic characteristics. This cleaned dataset is suitable for downstream tasks such as feature extraction, TF-IDF vectorization, and machine learning model training for emotion recognition.

Chapter 5

Data Visualization

5.1 Visualization Before Preprocessing

Figure 5.1 shows the distribution of emotion labels in the raw dataset before any cleaning or preprocessing. It highlights the imbalance among categories, with some emotions like *sadness* and *happiness* being the most frequent, while *joy* and *disgust* are underrepresented.

Detailed counts of the labels:

- Sadness: 1,112 samples
- Happiness: 897 samples
- Anger: 631 samples
- Neutral: 566 samples
- Fear: 375 samples
- Anxiety: 368 samples
- Joy: 77 samples
- Disgust: 67 samples

This visualization clearly shows the need for preprocessing and potential balancing techniques to improve model performance, especially for underrepresented classes.

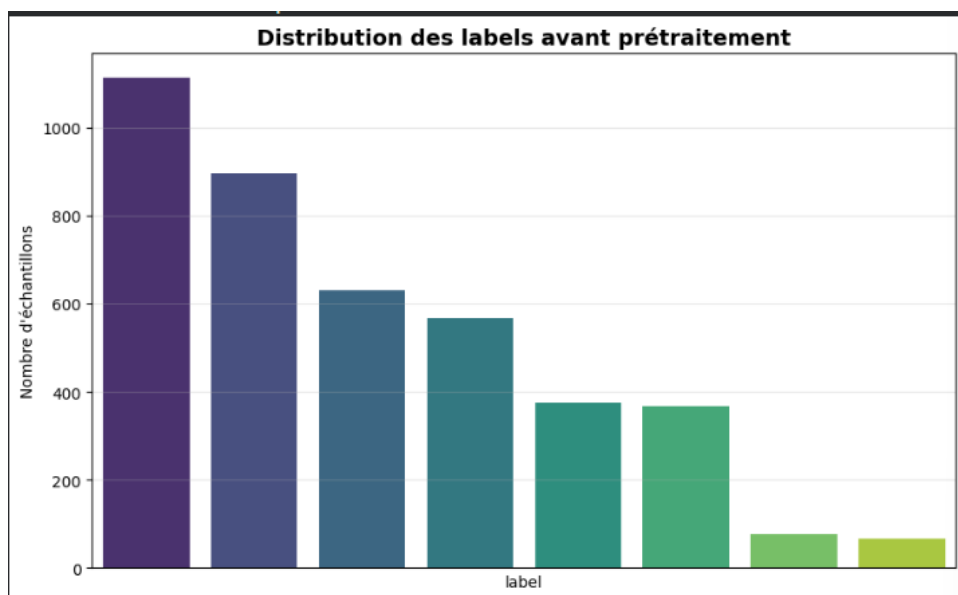


Figure 5.1: Distribution of emotion labels before preprocessing.

Chapter 6

References

- Scikit-learn documentation: <https://scikit-learn.org/stable/>
- Pandas documentation: <https://pandas.pydata.org/docs/>
- Matplotlib documentation: <https://matplotlib.org/stable/contents.html>
- Selenium documentation: <https://www.selenium.dev/documentation/>
- Reddit API reference: <https://www.reddit.com/dev/api/>
- Python official documentation: <https://docs.python.org/3/>